12 Respiratory system

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Introduction

Diseases of the respiratory system account for up to a third of deaths in most countries and for a major proportion of visits to the doctor and time away from work or school. As with every aspect of diagnosis in medicine, the key to success is a clear and carefully recorded history; symptoms may be trivial or extremely distressing, but either may indicate serious and life-threatening disease.

The history

Most patients with respiratory disease will present with breathlessness, cough, excess sputum, haemoptysis, wheeze or chest pain.

Breathlessness

Everyone becomes breathless on strenuous exertion. Breathlessness inappropriate to the level of physical exertion, or even occurring at rest, is called dyspnoea. Its mechanisms are complex and not fully understood. It is not caused simply by a lowered blood oxygen tension (hypoxia) or a raised blood carbon dioxide tension (hypercapnia), although these may play a significant part. People with cardiac disease (see Chapter 13) and even non-cardiorespiratory conditions, such as anaemia, thyrotoxicosis or metabolic acidosis, may become dyspnoeic as well as those with primarily respiratory problems (Box 12.1).

An important assessment is whether the dyspnoea is related only to exertion and how far the patient can walk at a normal pace on the level (exercise tolerance). This may take some skill to elicit, because few people note their symptoms in this form, but a brief discussion about what they can do in their daily lives usually gives a good estimate of their mobility (Box 12.2).

Other clarifications will include whether the symptoms vary, whether there are good days and bad days and, very importantly, whether there are any times of day or night that are usually worse than others. Variable airways obstruction caused by asthma is very often worse at night and in the early morning. By contrast, people with predominantly irreversible airways obstruction owing to chronic obstructive pulmonary disease (COPD) will often say that as long as they are sitting in bed, they feel quite normal; it is exercise that troubles them.

Cough

The symptom of cough can be short-lived or last years; cough can be defined as acute (lasting fewer than 3 weeks) or chronic (lasting more than 8 weeks) (Box 12.3). A cough may be dry or it may produce sputum. Acute cough is most commonly caused by recent infection, either viral or bacterial; however, any cough that is associated with haemoptysis should be a cause for concern, prompt appropriate assessment and a baseline chest X-ray (CXR) at the very least. Any patient with a chronic cough (i.e. one that lasts more than 8 weeks) should also be sent for a CXR and spirometry as baseline investigations (Box 12.4). Discussion about cough should include:

- How long has the cough been present? A cough lasting a few days following a cold has less significance than one lasting several weeks in a middle-aged smoker, which may be the first sign of a malignancy.
- Is the cough worse at any time of day or night? A dry cough at night may be an early symptom of asthma, as may a cough that comes in spasms lasting several minutes.
- Is the cough aggravated by anything, for example allergic triggers such as dust, animals or pollen, or non-specific triggers such as exercise or cold air? The increased reactivity of the airways seen in asthma and in some normal people for several weeks after viral respiratory infections may present in this way. Severe coughing, whatever its cause, may be followed by vomiting (Box 12.5).

Sputum

- Is sputum produced?
- What does it look like? Children and some adults swallow sputum, but it is always worth asking for a description of its colour and consistency. Yellow or green sputum is usually purulent. People with

Box 12.1 Causes of breathlessness

Acute	Subacute	Chronic
Airways obstruction	Pneumonia	COPD
Anaphylaxis	Exacerbation of COPD	Pleural effusion
Asthma	Angina	Malignancy
Pneumothorax	Cardiac tamponade	Chronic pulmonary emboli
Pulmonary embolus	Metabolic acidosis	Restrictive lung disorders, including interstitial lung disease
Myocardial infarction	Pain	Congestive cardiac failure
Pulmonary oedema	Pontine haemorrhage	Valvular dysfunction
Arrhythmias		Cardiomyopathy
Anxiety		Diastolic dysfunction
		Pulmonary hypertension
		Anaemia
		Neuromuscular disorders
		Deconditioning
		Obesity

COPD, chronic obstructive pulmonary disease.

Box 12.2 Medical Research Council grading of dyspnoea (breathlessness scale)

1. Not troubled by breathlessness except on strenuous exercise.
2. Short of breath when hurrying or walking up a slight hill.
3. Walks slower than contemporaries on the level because of breathlessness, or has to stop for breath when walking at own pace.
4. Stops for breath after about 100 m or after a few minutes on the level.
5. Too breathless to leave the house, or breathless when dressing or undressing.

asthma may produce small amounts of very thick or jelly-like sputum, sometimes in the shape of a cast of the airways. Eosinophils may accumulate in the sputum in asthma, causing a purulent appearance even when no infection is present.

- How much is produced? When severe lung damage in infancy and childhood was common, bronchiectasis was often found in adults. The amount of sputum produced daily often exceeded a cupful. Bronchiectasis is now rare, and chronic bronchitis causes the production of smaller amounts of sputum.

Box 12.3 Causes of cough

Causes of cough	Examples
Respiratory	Viral or bacterial infection, bronchospasm, COPD, non-asthmatic eosinophilic asthma, bronchiolitis, malignancy, parenchymal disease (e.g. ILD, bronchiectasis, cystic fibrosis, sarcoidosis, pleural disease, aspiration)
Upper airways disease	Post nasal drip, sinusitis, inhaled foreign body, tonsillar enlargement, laryngopharyngeal reflux (LPR)
Cardiovascular disease	LVF, mitral stenosis
Gastro-oesophageal disease	GORD
Neurological disease	Aspiration
Drugs and irritants	ACE inhibitors, cigarette smoke

ACE, angiotensin converting enzyme; COPD, chronic obstructive pulmonary disease; ILD, interstitial lung disease; LVF, left ventricular failure; GORD, gastro-oesophageal reflux disease—also associated with laryngopharyngeal reflux (LPR).

Box 12.4 Five most common causes of chronic cough with a normal CXR

- Post viral upper respiratory tract infection (URTI)
- Smoking
- Asthma—including cough variant asthma and non-asthmatic eosinophilic asthma
- Post nasal drip (hay fever)
- Gastro-oesophageal reflux disease (GORD)

Box 12.5 Important questions in the history of chronic cough

- Have you had a recent cold, sore throat or viral infection?
- Do you have a history of asthma, nocturnal cough or wheeze?
- Do you experience nasal discharge, hay fever or sinusitis?
- Do you suffer from acid reflux, indigestion or coughing after meals?
- What time of day is the cough worse?
- Do you smoke?
- Are you breathless?
- Have you coughed up blood?
- Do you have a hoarse voice?
- Have you had fevers or night sweats?
- Have you lost weight?
- Are you getting chest pain?

Haemoptysis

Haemoptysis is the coughing up of blood in the sputum. It should never be dismissed without very careful evaluation of the patient. The potentially serious significance of blood in the sputum is well

Box 12.6 Causes of haemoptysis

- Malignancy and benign lung tumours, including lung metastasis
- Pulmonary infection, including bacterial pneumonia, tuberculosis (TB), lung abscesses and fungal infection
- Bronchiectasis including cystic fibrosis
- Pulmonary emboli
- Congestive heart failure
- Pulmonary fibrosis
- Pulmonary vasculitis
- Severe pulmonary hypertension
- Arteriovenous malformation
- Chest trauma and foreign bodies
- Endometriosis
- Anticoagulation or coagulopathy
- Drugs (e.g. cocaine, thrombolytics)

known, and fear often leads patients not to mention it. A specific question is always necessary, as well as an attempt to decide if it is fresh or altered blood, how much is produced, when it started and how often it happens (Box 12.6).

Blood may be coughed up alone, or sputum may be bloodstained. It is sometimes difficult for the patient to describe whether the blood has originated from the chest or whether it comes from the gums or nose or even from the stomach. Always ask patients about associated conditions, such as epistaxis (nose bleeds) or the subsequent development of melaena (altered blood in the stool), which occurs in the case of upper gastrointestinal bleeding. Usually, however, it is clear that the blood originates from the chest, and this is an indication for further investigation.

Wheezing

Always ask whether the patient hears any noises coming from the chest. Even if a wheeze is not present when you examine the patient, it is useful to know that he has noticed it on occasions. Sometimes wheezing will have been noticed by others (especially by a partner at night, when asthma is worse) but not by the patient.

Sometimes stridor (see Chapter 22) may be mistaken for wheezing by both patient and doctor. This serious finding usually indicates narrowing of the larynx, trachea or main bronchi. Also, it is not unusual for patients with a pneumothorax to describe 'rubbing' or 'gurgling' sounds in their chest which may well be owing to the displaced lung.

Pain in the chest

Apart from musculoskeletal aches and pains consequent upon prolonged bouts of coughing, chest pain caused by lung disease usually arises from the pleura. Pleuritic pain is sharp and stabbing and is made worse by taking a deep breath or coughing. It occurs when the pleura is inflamed, most commonly by infection in the underlying lung. More constant pain, unrelated to breathing, may be caused by local invasion of the chest wall by a lung or pleural tumour.

A spontaneous pneumothorax causes pain which is worse on breathing, but which may have more of an aching character than the stabbing pain of pleurisy. If a pulmonary embolus causes infarction of the lung, pleurisy and hence pleuritic pain may occur, but an acute pulmonary embolus can also cause pain that is not stabbing in nature. A large pulmonary embolus causing haemodynamic disturbance may cause cardiac-type chest pain. Costocondritis and Tietze's syndrome can also cause acute pain and tenderness in the chest wall owing to inflammation around the chondrosternal joints.

Other symptoms

Quite apart from the common symptoms of respiratory disease, there are some other aspects of the history that are particularly relevant to the respiratory system.

Upper airway

Questions related to the ear, nose and throat are relevant. Rhinosinusitis often coexists with asthma or, less commonly, bronchiectasis and can be an aggravating factor. A common cause of chronic cough is postnasal drip secondary to rhinitis. A change in the voice may indicate involvement of the left recurrent laryngeal nerve by a carcinoma of the lung. Sometimes patients using inhaled corticosteroids for asthma develop oropharyngeal candidiasis or even hoarseness or weakness of the voice, which improves on changing the treatment. Laryngopharyngeal reflux (LPR) increasingly is recognized as a cause of hoarseness and chronic cough and can be diagnosed on direct visualization of the upper airway. However, do not ascribe hoarseness to these causes in older patients, because carcinoma of the vocal cords can also be present with hoarseness or a change in the quality of the voice. Laryngoscopy is always indicated if hoarseness persists for more than 4 weeks.

The smoking and recreational drug history

Always take a full smoking and recreational drug history. Do so in a sympathetic and non-judgemental way, or the detail is unlikely to be accurate. The time for advice about smoking cessation is after completion of your assessment, not at the outset. Simply asking 'Do you smoke?' is not enough. Novices will be astonished at how often closer probing of the answer 'no' reveals that the patient gave it up yesterday or that he states his intention of doing so from the time of your consultation. Age of starting and stopping, if an ex-smoker, and average consumption for both current and ex-smokers are the bare minimum information needed.

Identifying an individual as a current or ex-smoker will greatly influence the interpretation you place on your findings upon history and examination. Almost all cases of lung cancer and COPD occur in those who have smoked.

Recreational drug use tends to be more common in younger people, but do not assume that this is the case and ask all patients, of all ages, from all walks of life about such use. Again, sounding sympathetic rather than judgemental is crucial and a good opening line can be, 'If you don't mind me asking…'. Heroin, crack, cannabis and other drugs are smoked and, in some cases, cause more damage to the lungs than tobacco. Cannabis can cause severe emphysema in younger patients, who are often unaware of its effects. Use the consultation to discuss its long-term sequelae.

The family history

There is a strong inherited susceptibility to asthma. Associated atopic conditions, such as eczema and hay fever, may also be present in relatives of those with asthma, particularly in those who develop the condition when young.

The occupational history

No other organ is as susceptible to the working environment as are the lungs. Several hundred different substances have now been recognized as causing occupational asthma. Paint sprayers, workers in the electronics, rubber or plastics industries and woodworkers are relatively commonly affected (Box 12.7). Always ask about a relationship between symptoms and work.

Damage from inhalation of asbestos may take decades to become manifest, most seriously as malignant mesothelioma. In industrialized countries, this once extremely rare tumour of the pleura has become more common and will become even more so in the next 20 years. In middle-aged individuals who present with a pleural effusion (often the first sign of a mesothelioma) always ask about possible asbestos exposure in jobs back to the time of first employment (Box 12.8).

As far as the occupational history is concerned, the best way to proceed is chronologically. Most people cannot randomly remember, for example, what they might have been doing 20 years ago or indeed, if asked in isolation, when they worked in a particular job. However, if you start at the beginning of their life and work forward they find it much easier to remember (try it yourself starting with your school exams!)

Start by asking the patient how old he was when he left school, then what job or further education he had; then ask him to continue through his life to the present day. Particularly for those who went on to further education, ask about holiday jobs (you might be surprised at their responses!) and it might be worth asking if they travelled overseas with their employment, especially if they were in the armed forces. Do not assume that all 80-year-olds are retired or indeed that all young patients are employed.

Box 12.7 List of common occupations that may be associated with asthma

- Car paint sprayers—isocyanates
- Electricians—colophony
- Woodworkers
- Rubber and plastic industries
- Bakers—flour dust and enzymes, e.g. amylase
- Working with animals—vets, zoo keepers, laboratory worker—rodent urinary proteins
- Working with agriculture—farmers, fish worker—salmon proteins
- Health care professionals—latex and diathermy
- Hairdressing—persulphate, henna
- Tea sifters and packers

Box 12.8 List of activities that may lead to asbestos exposure

- Mining and manufacture of asbestos
- Shipbuilding and aircraft manufacturing
- Dock and rail workers—unloading asbestos from ships/trains
- Thermal and fire insulation—lagging
- Construction, building repair and demolition
- Plumbers and gas fitters
- Car mechanics (brake linings)
- Electricians, carpenters, upholsterers
- Manufacture of gas masks in World War II
- Teachers and health workers—old school buildings and hospitals as the source
- Family member of one of the above, and/or working or living near an asbestos source (particularly if asbestos fibres are taken home on workers' clothing)

The examination

General assessment

An examination of the respiratory system is incomplete without a simultaneous general assessment (Box 12.9). Watch the patient as he comes into the room, during your history taking and while he is undressing and climbing on to the couch. If this is a hospital inpatient, is there breathlessness just on moving in bed? Is he needing oxygen therapy? If so, at what flow rate and what is the mode of delivery—nasal cannula, Venturi mask, high flow? Look at the observation chart and review pulse, blood pressure, oxygen saturations and what is their early warning score (EWS)? A breathless patient may be using the accessory muscles of respiration (e.g. sternomastoid) and, in the presence of severe COPD, many patients find it easier to breathe out through pursed lips (Fig. 12.1).

Box 12.9 Points to note in a general assessment

- Physique and gait
- Voice
- Breathlessness
- Clubbing of the fingers
- Tobacco staining of fingers
- Bruising and/or thinness of skin
- Venous pulses
- Cyanosis or pallor
- Ptosis
- Swollen face
- Collateral vessels across anterior chest wall
- Intercostal recession
- Use of accessory respiratory muscles
- Lymph nodes



Figure 12.1 Respiratory failure. The patient is breathless at rest and there is central cyanosis with blueness of the lips and face. The lips are pursed during expiration, a characteristic feature of COPD. This facial appearance is often accompanied by heart failure with peripheral oedema (cor pulmonale).

Is there an audible wheeze or stridor?

- Is the voice hoarse?
- Is the patient continually coughing? Dry or productive?
- Is the patient capable of producing a normal, explosive cough, or is the voice weak or non-existent even when he is asked to cough?
- Is the wheezing audible, usually loudest in expiration, or is there stridor, a high-pitched inspiratory noise?
- What is on the bedside table (e.g. inhalers, a peak flow meter, tissues, a sputum pot, an oxygen mask, nebulizer, continuous positive airway pressure (CPAP) machine)?

Box 12.10 Signs to look for in the hands

- Clubbing
- Pallor
- Warm, well-perfused palms (CO_2 retention)
- Cyanosis
- Flap
- Tremor
- Tobacco staining
- Bruising and/or thin skin
- Pulse rate and character

Box 12.11 Observing the chest

- Rate of respiration
- Rhythm of respiration
- Chest expansion
- Symmetry
- Shape of the chest—elliptical, barrel, funnel
- Surgical scars

- What is the physique and state of general nourishment of the patient?

For the examination, the patient should be resting comfortably on a bed or couch, supported by pillows so that he can lean back comfortably at an angle of 45° (this is often more upright than patients choose for themselves).

Upper airways

A comprehensive account of examination of the upper airways is given in Chapter 22; a focused assessment should be included in the evaluation of all new patients or those with new upper airway symptoms.

Hands

The hands should be inspected for clubbing, pallor or cyanosis (Box 12.10). Tobacco-stained fingers may indicate a heavy smoker. Respiratory causes of clubbing include carcinoma of the bronchus, pulmonary fibrosis, bronchiectasis, lung abscess and pleural empyema. A fine tremor may indicate the use of inhaled β_2 agonists, such as salbutamol. A flap may indicate carbon dioxide retention or hypercapnia. Such patients are often drowsy, with warm hands and a bounding pulse. In a significant asthma attack, the pulse rate is usually raised. The systolic blood pressure also falls during the severe inspiratory effort of acute asthma, and the degree of this fall (the degree of pulsus paradoxus) can be used as a measure of asthma severity.

Respiratory rate and rhythm

The respiratory rate and pattern of respiration should be noted. The normal rate of respiration in a relaxed adult is about 14–16 breaths per minute (Box 12.11). Tachypnoea is an increased respiratory rate observed by the doctor, whereas dyspnoea is

the symptom of breathlessness experienced by the patient. Apnoea means cessation of respiration.

Cheyne-Stokes breathing is the name given to a disturbance of respiratory rhythm in which there is cyclical deepening and quickening of respiration, followed by diminishing respiratory effort and rate, sometimes associated with a short period of complete apnoea, the cycle then being repeated. This is often observed in severely ill patients and particularly in severe cardiac failure, narcotic drug poisoning and neurological disorders. It is occasionally seen, especially during sleep, in elderly patients without any obvious serious disease.

Some patients may have apnoeic episodes during sleep owing to complete cessation of respiratory effort (central apnoea) or, much more commonly, apnoea despite continuation of respiratory effort. This, which is known as obstructive sleep apnoea, is caused by obstruction of the upper airways by soft tissues in the region of the pharynx and is most common in obese patients.

Venous pulses

The venous pulses in the neck (see Chapter 13) should be inspected. A raised jugular venous pressure (JVP) may be a sign of cor pulmonale, right heart failure caused by chronic pulmonary hypertension in severe lung disease, commonly COPD. Pitting oedema of the ankles and sacrum is usually present. However, engorged neck veins can be caused by superior vena cava obstruction (SVCO), usually because of malignancy in the upper mediastinum. SVCO can also be associated with facial swelling and plethora (redness) and collateral circulation across the anterior chest wall.

Head

Examination of the eyes may reveal anaemia or, rarely, Horner's syndrome, secondary to a cancer at the lung apex (Pancoast tumour) invading the cervical sympathetic chain. The lips and tongue should be inspected for central cyanosis, which almost always indicates poor oxygenation of the blood by the lungs, whereas peripheral cyanosis alone is usually owing to poor peripheral perfusion. Oral candida may indicate use of inhaled steroids or be a sign of debilitation or underlying immune suppression in the patient.

Examination of the chest

Relevant anatomy

The interpretation of signs in the chest often causes problems for the beginner. A review of the relevant anatomy may help.

The bifurcation of the trachea corresponds on the anterior chest wall with the sternal angle, the transverse bony ridge at the junction of the body of the sternum and the manubrium sterni. Posteriorly, the level is at the disc between the fourth and fifth thoracic vertebrae. The ribs are most easily counted downwards from the second costal cartilage, which articulates with the sternum at the extremity of the sternal angle.

A line from the second thoracic spine to the sixth rib, in line with the nipple, corresponds to the upper border of the lower lobe (oblique or major interlobar fissure). On the right side, a horizontal line from the sternum at the level of the fourth costal cartilage, drawn to meet the line of the major interlobar fissure, marks the boundary between the upper and middle lobes (the horizontal or minor interlobar fissure). The greater part of each lung, as seen from behind, is composed of the lower lobe; only the apex belongs to the upper lobe. The middle and upper lobes on the right side and the upper lobe on the left side occupy most of the area in front (Fig. 12.2). This is most easily visualized if the lobes are thought of as two wedges fitting together, not as two cubes piled one on top of the other (Fig. 12.3).

The stethoscope is so much a part of the 'image' of a doctor that it is very easy for the student to forget that listening is only one part of the chest examination. Obtaining the maximum possible information from your examination requires you to look, then to feel and, only then, to listen.

Looking: inspection of the chest

Appearance of the chest

First, look for any obvious scars from previous surgery. Thoracotomy scars (from lobectomy or pneumonectomy (removal of the whole lung)) are usually visible running from below the scapula posteriorly, sweeping round the axilla to the anterior chest wall. Pleural procedures, such as intercostal drain insertion, biopsy or VATS (video-assisted thoracoscopic surgery), may be associated with small scars, often in the axilla or posteriorly. A small scar above the sternal notch indicates a previous tracheostomy. Older patients may have small scars in the midline below the clavicle indicative of a phrenic nerve crush (a previous treatment for TB). Look for any lumps visible beneath the skin or any lesions on the skin itself. If you are examining from the right of the patient, ensure that you thoroughly inspect the left side. It is easy to miss a lateral thoracotomy scar or one that is hidden in a skinfold.

Next, inspect the shape of the chest itself. The normal chest is bilaterally symmetrical and elliptical in horizontal cross-section, with the narrower diameter being anteroposterior. The chest may be distorted by disease of the ribs or spinal vertebrae as well as by underlying lung disease (Box 12.12). Lobar collapse produces characteristic changes on chest X-ray and they are shown in Figure 12.4.

Kyphosis (forward bending) or scoliosis (lateral bending) of the vertebral column will lead to asymmetry of the chest and, if severe, may significantly restrict lung movement. A normal chest X-ray is

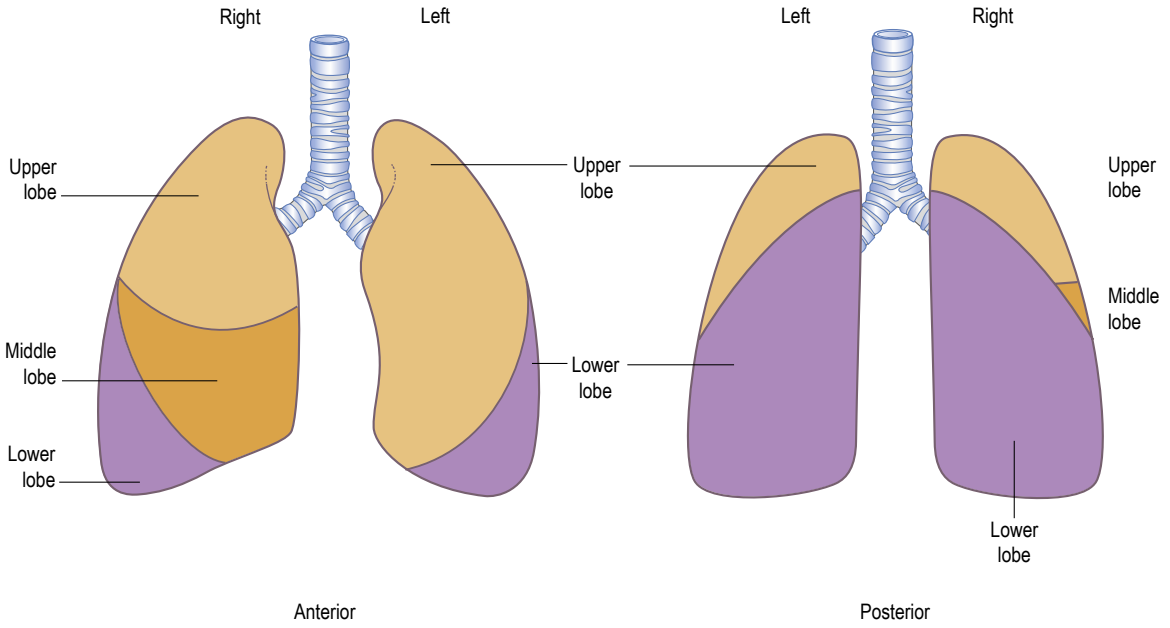


Figure 12.2 Anterior and posterior aspects of the lungs.

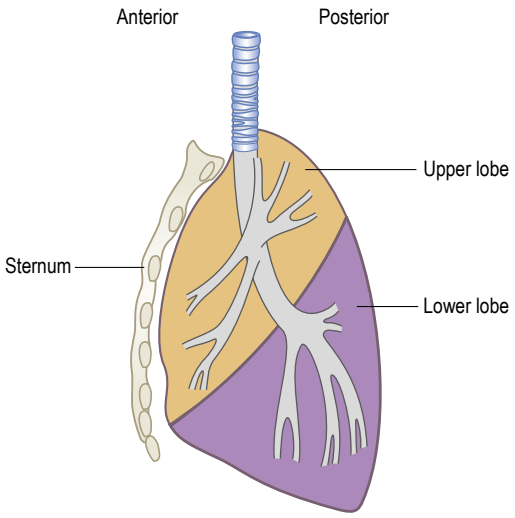


Figure 12.3 Lateral aspect of the left lung.

seen in Figure 12.5. Severe airways obstruction, particularly long-term as in COPD (Fig. 12.6), may lead to overinflated lungs. On examination, the chest may be 'barrel shaped', most easily appreciated as an increased anteroposterior diameter, making the horizontal cross-section more circular. On X-ray, the hemidiaphragms appear lower than usual, and flattened.

Movement of the chest

Look to see if the chest movements are symmetrical. If they seem to be diminished on one side, then that is likely to be the side on which is an abnormality. Intercostal recession, a drawing-in of the intercostal spaces with inspiration, may indicate severe upper airways obstruction, as in laryngeal disease or tumours of the trachea. In COPD, the lower ribs often move paradoxically inwards on inspiration instead of the normal outwards movement.

Box 12.12 Features to note in assessing the shape of the chest

- Kyphosis
- Scoliosis
- Flattening
- Pectus excavatum, pectus carinatum
- Overinflation
- Previous surgery causing asymmetry, such as thoracoplasty

Feeling: palpation of the chest

Lymph nodes

The lymph nodes in the supraclavicular fossae, cervical regions and axillary regions should be palpated; do not forget to feel gently behind the sternocleidomastoid muscles. If they are enlarged, this may be secondary to the spread of malignant disease from the chest, and such findings will influence decisions regarding treatment. Lymph nodes in the neck are best felt by sitting the patient up and examining from behind.

Swellings and tenderness

It is useful to palpate any part of the chest that presents an obvious swelling or where the patient complains of pain (Box 12.13). Feel gently, because pressure may increase the pain. It is often important, particularly in the case of musculoskeletal pain, to identify a site of tenderness (Box 12.14). Surgical emphysema (air

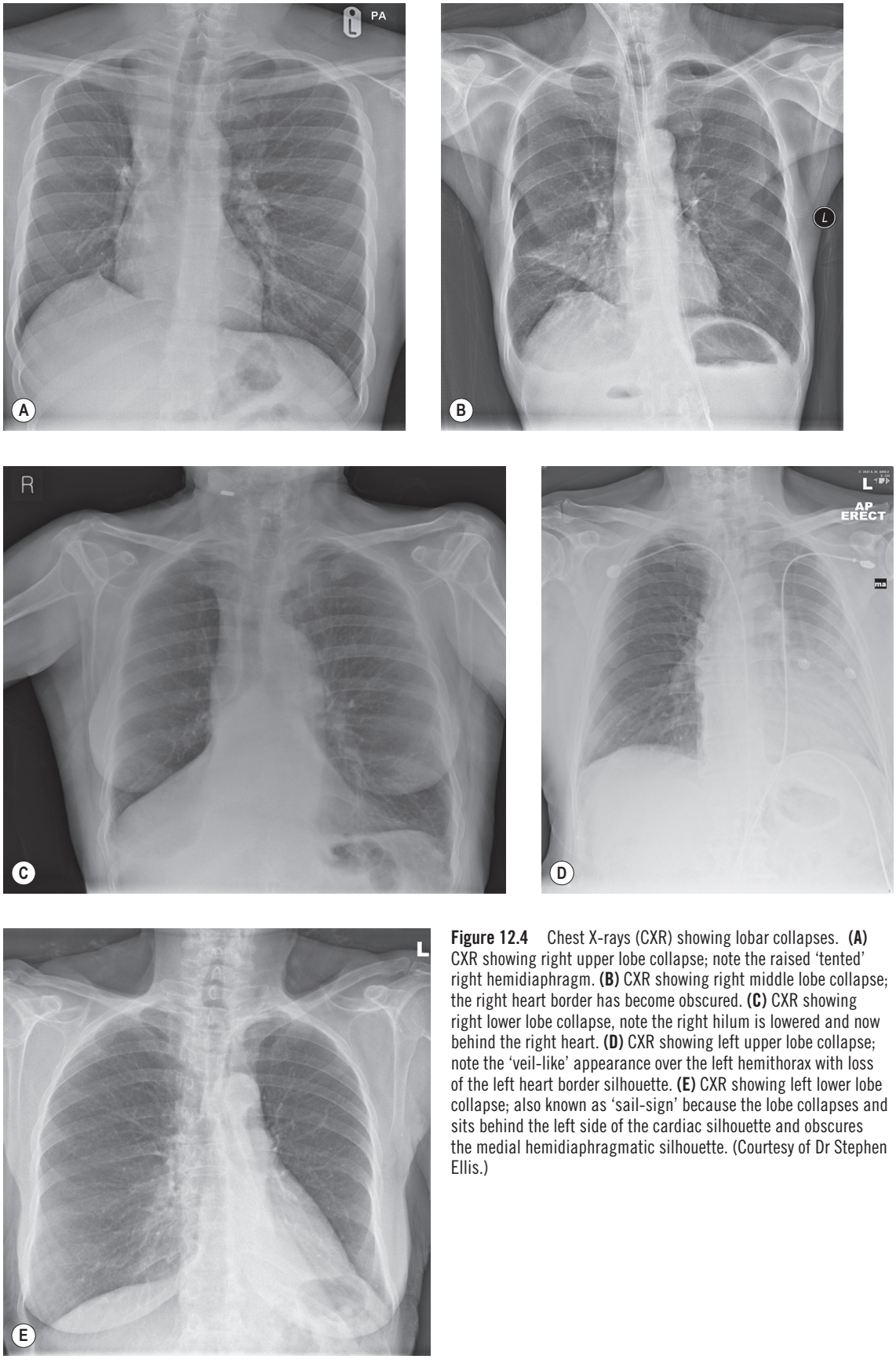


Figure 12.4 Chest X-rays (CXR) showing lobar collapses. **(A)** CXR showing right upper lobe collapse; note the raised 'tented' right hemidiaphragm. **(B)** CXR showing right middle lobe collapse; the right heart border has become obscured. **(C)** CXR showing right lower lobe collapse, note the right hilum is lowered and now behind the right heart. **(D)** CXR showing left upper lobe collapse; note the 'veil-like' appearance over the left hemithorax with loss of the left heart border silhouette. **(E)** CXR showing left lower lobe collapse; also known as 'sail-sign' because the lobe collapses and sits behind the left side of the cardiac silhouette and obscures the medial hemidiaphragmatic silhouette. (Courtesy of Dr Stephen Ellis.)

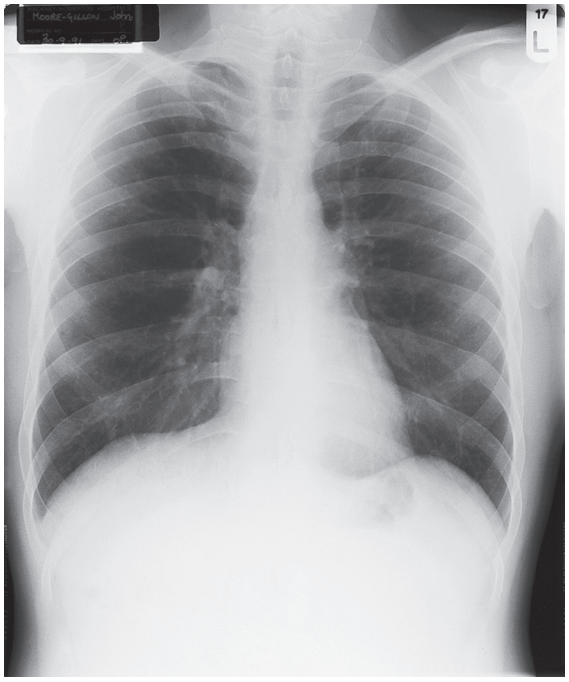


Figure 12.5 Normal chest X-ray.

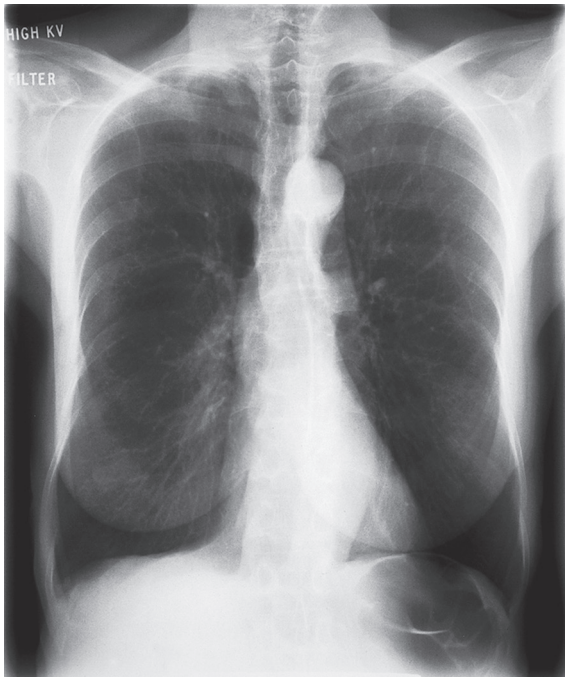


Figure 12.6 Chest X-ray in severe chronic obstructive pulmonary disease.

in the tissues), which feels like popcorn or bubble paper underneath the skin, is caused by trauma, pneumothorax, pneumomediastinum and infection, as well as chest instrumentation following surgery or a chest drain.

Trachea and heart

The positions of the cardiac impulse and trachea should then be determined. Feel for the trachea by putting the second and fourth fingers of the examining hand on each edge of the sternal notch and use the third finger to assess whether the trachea is central or deviated to one side. Warn the patient in advance what you are about to do and avoid heavy-handedness in this situation. Rough technique is uncomfortable for the patient who may feel like he is being choked. A slight deviation of the trachea to the right may be found in healthy people.

Displacement of the cardiac impulse without displacement of the trachea may be owing to scoliosis, to a congenital funnel depression of the sternum (pectus excavatum) or to enlargement of the left ventricle. In the absence of these conditions, a significant displacement of the cardiac impulse or trachea or of both together suggests that the position of the mediastinum has been altered by disease of the lungs or pleura. The mediastinum may be pushed away from the affected side (contralateral deviation) by a pleural effusion or pneumothorax. Fibrosis or collapse of the lung will pull the mediastinum towards the affected side (ipsilateral deviation).

Box 12.13 Points to note on palpation of the chest

- Swelling
- Surgical emphysema
- Pain and tenderness, including over the spine
- Tracheal position
- Cardiac impulse
- Asymmetry
- Tactile vocal fremitus

Box 12.14 Causes of pain and tenderness in the chest

- A recent injury of the chest wall or inflammatory conditions, cough fracture
- Intercostal muscular pain—as a rule, localized painful spots can be discovered on pressure
- A painful costochondral junction (e.g. Tietze's syndrome)
- Secondary malignant deposits in the rib
- Herpes zoster before the appearance of the rash
- Spinal tenderness secondary to infection or malignancy

Chest expansion

As well as by simple inspection, possible asymmetrical expansion of the chest may be explored further by palpation. Face the patient and place the fingertips of both hands on either side of the lower ribcage so that the tips of the thumbs meet in the midline in front of the chest but are not touching the skin. A deep breath by the patient will increase the distance between the thumbs and indicate the degree of expansion. If one thumb remains closer to the midline, this suggests diminished expansion on that side. Essentially the hands are being used like a pair of calipers to measure expansion in the lateral bases of the lungs where maximum expansion occurs.

Tactile vocal fremitus is detected by palpation, but this is not a commonly used routine examination technique. It is discussed further under auscultation, below.

Feeling: percussion of the chest

The technique of percussion was probably developed as a way of ascertaining how much fluid remained in barrels of wine or other liquids. Auenbrugger applied percussion to the chest, having learned this method in his father's wine cellar. Effective percussion is a knack that requires consistent practice; do so upon yourself or on willing colleagues, because percussion can be uncomfortable for patients if performed repeatedly and inexpertly.

The middle finger of the left hand is placed on the part to be percussed and pressed firmly against it, with slight hyperextension of the distal interphalangeal joint. The back of this joint is then struck with the tip of the middle finger of the right hand (vice versa if you are left-handed). The movement should be at your wrist rather than at your elbow. The percussing finger is bent so that its terminal phalanx is at right angles and it strikes the other finger perpendicularly. As soon as the blow has been given, the striking finger is raised: the action is a tapping movement.

The two most common mistakes made by the beginner are first, failing to ensure that the finger of the left hand is applied flatly and firmly to the chest wall, with slight pressure, and second, striking the percussion blow from the elbow rather than from the wrist. The character of the sound produced varies both qualitatively and quantitatively (Box 12.15). When the air in a cavity of sufficient size and appropriate shape is set vibrating, a resonant sound is produced, and a characteristic sensation is also felt by the finger placed on the chest. Try tapping a hollow cupboard and then a solid wall. The feeling is different as well as the sound. The sound and feel of resonance over a healthy lung are learned by practice, and it is against this standard that possible abnormalities of percussion must be judged.

The normal degree of resonance varies between individuals and in different parts of the chest in the same individual, being most resonant below the clavicles anteriorly and the scapulae posteriorly where the muscles are relatively thin and least resonant over the scapulae. On the right side, there is loss of resonance inferiorly as the liver is encountered. On the left side, the lower border overlaps the stomach, so there is a transition from lung resonance to tympanitic stomach resonance.

Always systematically compare the percussion note on the two sides of the chest, moving backwards and forwards from one side to the other, not all the way down one side and then down the other. Percuss over the clavicles; traditionally, this is done without an intervening finger on the chest, but there is no reason for this and it is more comfortable for the patient if the finger of the left hand is used in the usual way. Percuss three or four areas on the anterior chest wall, comparing left with right. Percuss the axillae, then three or four areas on the back of the chest.

Box 12.15 Points to note on percussion of the chest

- Resonance
- Dullness
- Pain and tenderness

Reduction of resonance (i.e. the percussion note is said to be dull) occurs in two important circumstances:

1. When the underlying lung is more solid than usual, usually because of consolidation or collapse
2. When the pleural cavity contains fluid (i.e. a pleural effusion is present).

Less commonly, a dull percussion note may be owing to thickened pleura. The percussion note is most dull when there is underlying fluid, as in a pleural effusion. Pleural effusion causes the sensation in the percussed finger to be similar to that felt when a solid wall is percussed. This is often called 'stony dullness'. By comparing side with side, it is usually easy to detect a unilateral pleural effusion. Pleural effusion usually leads to decreased chest wall movement. Effusions may occur bilaterally in some patients, and this may be more difficult to detect clinically.

An increase in resonance, or hyper-resonance, is more difficult to detect than dullness, and there is no absolute level of normal percussion against which extra resonance can be judged. It may be noticeable when the pleural cavity contains air, as in pneumothorax. Sometimes, however, in this situation one is tempted to think that the slightly duller side is the abnormal side. Further examination and chest X-ray will reveal the true situation, remembering that the side of the chest that expands less is usually the side with pathology.

Listening: auscultation of the chest

Listen to the chest with the diaphragm, not the bell, of the stethoscope (chest sounds are relatively high pitched, and therefore the diaphragm is more sensitive than the bell). Ask the patient to take deep breaths in and out through the mouth. Demonstrate what you would like the patient to do, and then check visually that he is doing it while you listen to the chest. If the patient has a tendency to cough, ask him to breathe more deeply than usual but not so much as to induce a cough with each breath. As with percussion, you should listen in comparable positions to each side alternately, switching back and forth from one side to the other to compare (Box 12.16).

Box 12.16 Points to note on auscultation of the chest

- Vesicular breath sounds—normal breath sounds
- Bronchial breath sounds—consolidation
- Vocal fremitus and resonance:
  - whispering pectoriloquy—consolidation
  - aegophony—top of pleural effusion, consolidation
- Added sounds:
  - pleural rub—associated with infection
  - wheezes—asthma, chronic obstructive pulmonary disease (COPD), infection, cardiac failure
  - crackles—pulmonary fibrosis, cardiac failure, COPD

The breath sounds

Breath sounds have intensity and quality. The intensity (or loudness) of the sounds may be normal, reduced or increased. The quality of normal breath sounds is described as vesicular.

Breath sounds will be normal in intensity when the lung is inflating normally, but may be reduced if there is localized airway narrowing, if the lung is extensively damaged by a process such as emphysema or if there is intervening pleural thickening or pleural fluid. Breath sounds may be of increased intensity in very thin subjects.

Breath sounds probably originate from turbulent airflow in the larger airways. When you place your stethoscope upon the chest, you are listening to how those sounds have been changed on their journey from their site of origin to the position of your stethoscope diaphragm. Normal lung tissue makes the sound quieter and selectively filters out some of the higher frequencies. The resulting sound that you hear is called a vesicular breath sound. There is usually no distinct pause between the end of inspiration and the beginning of expiration.

When the area underlying the stethoscope is airless, as in consolidation, the sounds generated in the large airways are transmitted more efficiently, so they are louder and there is less filtering of the high frequencies. The resulting sounds heard by the stethoscope are termed bronchial breathing, classically heard over an area of consolidated lung in cases of pneumonia. The sound resembles that obtained by listening over the trachea, although the noise there is much louder. The quality of the sound is rather harsh, the higher frequencies being heard more clearly. The expiratory sound has a more sibilant (hissing) character than the inspiratory one and lasts for most of the expiratory phase.

The intensity and quality of all breath sounds is so variable from patient to patient and in different situations that it is only by repeated auscultation of the chests of many patients that one becomes familiar with the normal variations and learns to recognize the abnormalities.

Added sounds

Added sounds are abnormal sounds that arise in the lung itself or in the pleura. The added sounds most commonly arising in the lung are best referred to as wheezes and crackles. Older terms such as râles to describe coarse crackles, crepitations to describe fine crackles and rhonchi to describe wheezes are poorly defined, have led to confusion and are best avoided.

Wheezes are musical sounds associated with airway narrowing. Widespread polyphonic wheezes, particularly heard in expiration, are the most common ones; they are characteristic of diffuse airflow obstruction, especially in asthma and COPD. These wheezes are probably related to dynamic compression of the bronchi, which is accentuated in expiration when airway narrowing is present. A fixed monophonic wheeze can be generated by localized narrowing of a single bronchus, as may occur in the presence of a tumour or foreign body. It may be inspiratory or expiratory or both and may change its intensity in different positions.

Wheezing generated in smaller airways should not be mistaken for stridor associated with laryngeal disease or localized narrowing of the trachea or the large airways. Stridor almost always indicates a serious condition requiring urgent investigation and management. The noise is often both inspiratory and expiratory. It may be heard at the open mouth without the aid of the stethoscope. On auscultation of the chest, stridor is usually loudest over the trachea.

Crackles are short, explosive sounds often described as bubbling or clicking. When the large airways are full of sputum, a coarse, rattling sound may be heard even without the stethoscope. However, crackles are not usually produced by moistness in the lungs. It is more likely that they are produced by sudden changes in gas pressure related to the sudden opening of previously closed small airways. Crackles at the beginning of inspiration are common in patients with COPD. Localized loud and coarse crackles may indicate an area of bronchiectasis. Crackles are also heard in pulmonary oedema. In diffuse interstitial fibrosis, crackles are characteristically fine in character and late inspiratory in timing (and said to sound like rolling your fingers through your hair near your ear).

The pleural rub, which is characteristic of pleural inflammation, usually occurs in association with pleuritic pain. It has a creaking or rubbing characteristic (said to sound like a foot crunching through fresh-fallen snow) and, in some instances, can be felt with the palpating hand as well as being audible with the stethoscope.

Take care to exclude false added sounds. Sounds resembling pleural rubs may be produced by movement of the stethoscope on the patient's skin or of clothes against the stethoscope tubing. Sounds arising in the patient's muscles may resemble added sounds: in particular, the shivering of a cold patient makes any attempt at auscultation almost useless. The stethoscope rubbing over hairy skin may produce sounds that resemble fine crackles.

Vocal resonance

When listening to the breath sounds, you are detecting with the stethoscope vibrations that have been made in the large airways. Vocal resonance is the resonance within the chest of sounds made by the voice. Vocal resonance is the detection of vibrations transmitted to the chest from the vocal cords as the patient repeats a phrase, usually the words 'ninety-nine'. The ear perceives not the distinct syllables but a resonant sound, the intensity of which depends on the loudness and depth of the patient's voice and the conductivity of the lungs. As always in examining the chest, each point examined on one side should be compared at once with the corresponding point on the other side.

Not surprisingly, conditions that increase or reduce conduction of breath sounds to the stethoscope have similar effects on vocal resonance. Consolidated lung conducts sounds better than air-containing lung, so in consolidation the vocal resonance is increased and the sounds are louder and often clearer. In such circumstances, even when the patient whispers a phrase (e.g. 'one, two, three'), the sounds may be heard clearly; this is known as whispering pectoriloquy. Above the level of a pleural effusion, or in some cases over an area of consolidation, the voice may sound nasal or bleating; this is known as aegophony, but is an unusual physical finding.

Vocal fremitus

Vocal fremitus is detected with the hand on the chest wall. It should, therefore, perhaps be regarded as part of palpation, but it is usually carried out after auscultation (see below). As with vocal resonance, the patient is asked to repeat a phrase such as 'ninety-nine'. The examining hand feels distinct vibrations when this is done. Some examiners use the ulnar border of the hand, but there is no good reason for this; the flat of the hand, including the fingertips, is far more sensitive, but make sure the fingers are pressed firmly onto the chest wall.

From the above, it should be clear that listening to the breath sounds, listening to the vocal resonance and eliciting vocal fremitus that they are all doing essentially the same thing: they are investigating how vibrations generated in the larynx or large airways are transmitted to the examining instrument, the stethoscope in the first two cases and the fingers in the third. It follows that in the various pathological situations, all three physical signs should behave in similar ways. Where there is consolidation, the breath sounds are better transmitted to the stethoscope, so they are louder and there is less attenuation of the higher frequencies, that is, 'bronchial breathing' is heard. Similarly, the vocal resonance and the vocal fremitus are increased. Where there is a pleural effusion, the breath sounds are quieter or absent and the vocal resonance and vocal fremitus are reduced or absent.

The intelligent student should now ask: 'Why try to elicit all three signs?' The experienced physician will answer: 'Because it is often difficult to interpret the signs that have been elicited, and three pieces of information are more reliable than one'.

Putting it together: an examination of the chest

There is no single perfect way of examining the chest, and most doctors develop their own minor variations of order and procedure. The following is one scheme that combines efficiency with thoroughness:

- Observe the patient generally and the surroundings. Look for any medicine, sputum pots, inhalers, nebulizers or, for example, CPAP machine around the patient's bed. Look at the patient's observation chart. Is the patient using oxygen; if so, how much, what is the rate?
- Ask the patient's permission for the examination and ensure he is lying comfortably at 45°.
- Examine the hands and take the pulse.
- Count the respiratory rate.
- Assess the JVP.
- Check the face for signs of anaemia or cyanosis as well as evidence of ptosis and miosis.
- Inspect the chest movements and the anterior chest wall. What is the shape of the chest?
- Feel the position of the trachea and check for axillary lymphadenopathy.
- Feel the position of the apex beat.
- Check the symmetry of the chest movements by palpation.
- Percuss the anterior chest and axillae.

Sit the patient forward:

- Inspect the posterior chest wall.
- Check for cervical and supraclavicular lymphadenopathy.
- Percuss the back of the chest.
- Listen to the breath sounds.
- Check the vocal resonance.
- Check the tactile vocal fremitus.
- Check for sacral oedema.

If you are examining a hospital inpatient, always take the opportunity to turn the pillow over before lying the patient back again; a cool, freshened pillow is a great comfort to an ill person.

- Listen to the breath sounds on the front of the chest.
- Check the vocal resonance.
- Check the tactile vocal fremitus.
- Check for pitting oedema of the ankles.

Stand back for a moment and reflect upon whether you have omitted anything or whether you need to check or repeat anything. Thank the patient and ensure he is dressed or appropriately covered.

Putting it together: interpreting the signs

Developing an appropriate differential diagnosis on the basis of the signs you have elicited requires thought and practice. Keeping the following in mind will help:

Box 12.17 Characteristics to note when assessing sputum

- Mucoid
- Purulent
- Frothy
- Bloodstained
- Rusty
- Frank haemoptysis
- Casts

- If movements are diminished on one side, there is likely to be an abnormality on that side.
- The percussion note is dull over a pleural effusion and over an area of consolidation; the duller the note, the more likely it is to be a pleural effusion.
- The breath sounds, the vocal resonance and the tactile vocal fremitus are quieter or less obvious over a pleural effusion, and louder or more obvious over an area of consolidation.
- Over a pneumothorax, the percussion note is more resonant than normal, but the breath sounds, vocal resonance and tactile vocal fremitus are quieter or reduced. Pneumothorax is easily missed.

Other investigations

Sputum examination

At the bedside

Hospital inpatients should have a sputum pot which must be inspected (Box 12.17). Mucoid sputum is characteristic in patients with chronic bronchitis when there is no active infection. It is clear and sticky and not necessarily produced in a large volume. Sputum may become mucopurulent or purulent when bacterial infection is present in patients with bronchitis, pneumonia, bronchiectasis or a lung abscess. In these last two conditions, the quantities may be large and the sputum is often foul smelling.

Occasionally asthmatics have a yellow tinge to the sputum, owing to the presence of many eosinophils. People with asthma may also produce a particularly tenacious form of mucoid sputum, and sometimes they cough up casts of the bronchial tree, particularly after an attack. Patients with bronchopulmonary aspergillosis may bring up black sputum or sputum with black parts in it, which is the fungal element of the *Aspergillus*.

When sputum is particularly foul smelling, the presence of anaerobic organisms should be suspected. Very ill patients with pulmonary oedema may bring up pink or white frothy sputum. Rusty-coloured sputum is characteristic of pneumococcal lobar pneumonia. Blood may be coughed up alone or bloodstained sputum produced in bronchogenic carcinoma, pulmonary tuberculosis (TB), pulmonary embolism, bronchiectasis or pulmonary hypertension (e.g. with mitral stenosis) being possible causes.

In the laboratory

Sputum may be examined under the microscope in the laboratory for the presence of pus cells and organisms and may be cultured in an attempt to identify the causative agent of an infection and antibiotic resistance patterns. It is seldom practical to wait for the results of such examinations, and most clinical decisions have to be based on the clinical probability of a particular infection being present.

Do not forget to ask for sputum to be examined for acid-fast bacilli when appropriate; tuberculosis requires specialized techniques of laboratory microscopy and culture to identify the responsible organisms; if the diagnosis is suspected, these tests must be specifically requested. Non-tuberculous mycobacteria (NTN) can occur in patients with chronic underlying lung pathology such as COPD and bronchiectasis.

Lung function tests

Measurements of respiratory function may provide valuable information. First, in conjunction with the clinical assessment and other investigations, they may help establish a diagnosis. Second, they will help indicate the severity of the condition. Third, serial measurements over time will show changes indicating disease progression or, alternatively, a favourable response to treatment. Finally, regular monitoring of lung function in chronic diseases, such as idiopathic pulmonary fibrosis, cystic fibrosis or obstructive airways disease, may warn of deterioration.

Simple respiratory function tests fall into three main groups:

1. Measuring the size of the lungs
2. Measuring how easily air flows into and out of the airways
3. Measuring how efficient the lungs are in the process of gas exchange

A spirometer will measure how much air can be exhaled after a maximal inspiration: the patient breathes in as much as he can, then blows out into the spirometer until no more air at all can be breathed out. This volume is called the vital capacity (VC). The amount of air in the lungs at full inspiration is a measure of the total lung capacity and that still remaining after a full expiration is called the residual volume (RV).

The actual value of total lung capacity (TLC) cannot be measured with a spirometer. The simplest way of determining it is to get the patient to inspire a known volume of air containing a known concentration of helium. Measuring the new concentration of helium that exists after mixing with the air already in the lungs enables the total lung capacity (TLC) to be calculated. Subtraction of the VC from this value gives the RV.

Usually, vital capacity is measured after the patient has blown as hard and fast as possible into

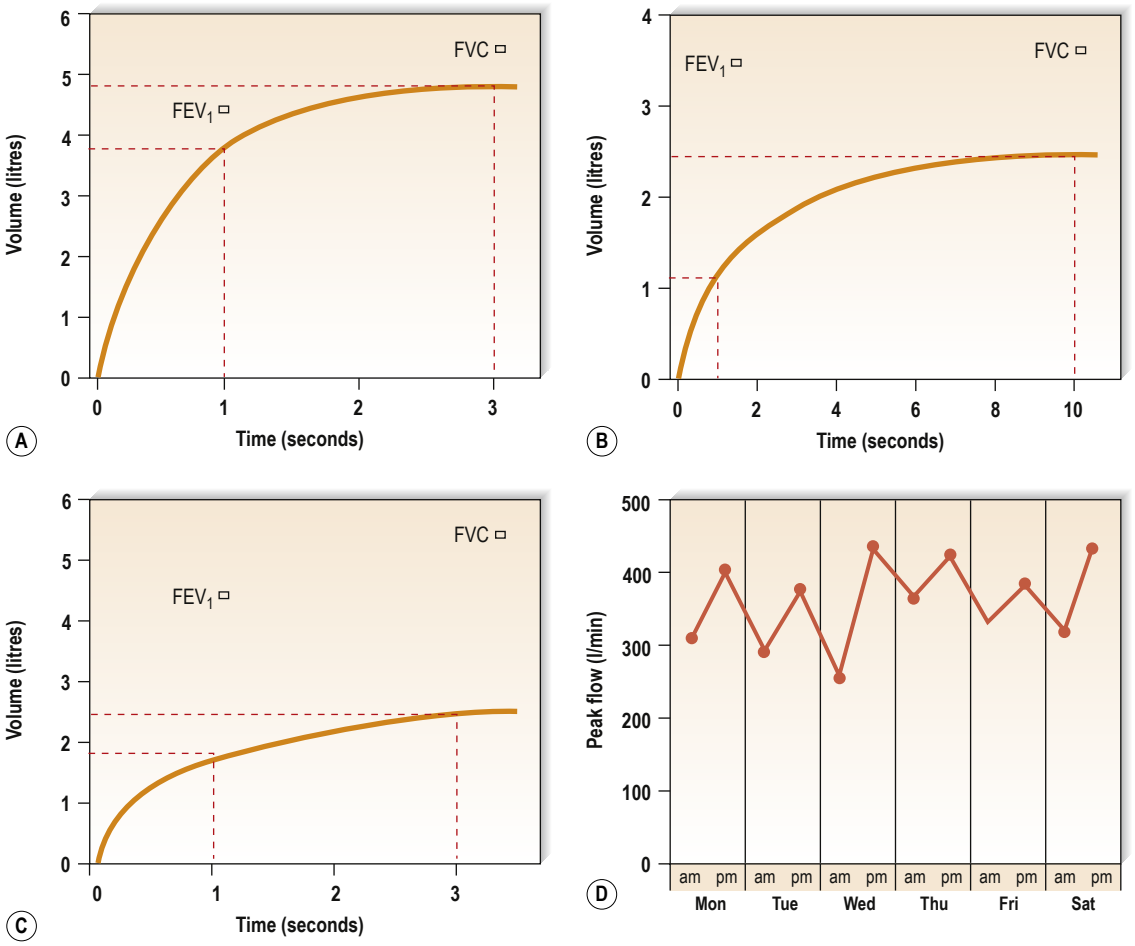


Figure 12.7 **(A)** Normal expiratory spirometer trace. **(B)** Spirometer trace showing an obstructive defect. Note the very prolonged (10-second) expiration. **(C)** Spirometer trace showing a restrictive defect. **(D)** Typical diurnal variation of peak flow, worse in the mornings, seen in a young asthmatic, during an exacerbation.

the spirometer, when the measurement is known as the forced vital capacity, or FVC. In normal lungs, VC and FVC are almost identical, but in COPD, compression of the airways during a forced expiration leads to closure of the airways earlier than usual, and FVC may be less than VC.

Figure 12.7A shows the trace produced by a spirometer. Time in seconds is on the *x*-axis and volume in litres is on the *y*-axis. Thus, the trace moves up during expiration assessing FVC and along the *x*-axis as time passes during expiration.

The volume of air breathed out in the first second of a forced expiration is known as the forced expiratory volume in the first second—almost always abbreviated to FEV_1. In normal lungs, the FEV_1 is more than 70% of FVC. When there is obstruction to airflow, as in COPD, the time taken to expire fully is prolonged and the ratio of FEV_1 to FVC is reduced. An example is shown in Figure 12.7B. A trace like this is described as showing an obstructive ventilatory defect. As noted above, the FVC may be reduced in severe airways obstruction but, in such cases, the FEV_1 is reduced even more and the FEV_1-to-FVC ratio remains low.

Some lung conditions restrict expansion of the lungs but do not interfere with the airways. In such individuals, both FEV_1 and FVC are reduced in proportion to each other, so the ratio remains normal even though the absolute values are reduced. Figure 12.7C shows a trace of this kind, a restrictive ventilatory defect in a patient with diffuse pulmonary fibrosis.

Look again at the normal expiratory spirogram (Fig. 12.7A). The slope of the trace is steepest at the onset of expiration. The trace thus shows that the rate of volume change with time is greatest in early expiration; in other words, the rate of airflow is greatest then. This measurement, the peak expiratory flow rate (PEFR), can be measured easily with a peak flow meter. A simplified version of this device is shown in Figure 12.8. This mini-peak flow meter is light and inexpensive. People with asthma can use it to monitor themselves and alter their medication, as suggested by their doctor, at the first signs of any



Figure 12.8 A mini-peak flow meter.

fall in peak flow measurement, which indicates a deterioration in their condition (Fig. 12.7D).

Normal gas exchange consists of the uptake of oxygen into the pulmonary capillary blood and the release of carbon dioxide into the alveoli. For this to be achieved, the ventilation of the lungs by air and their perfusion by blood need to be matched anatomically. An approximation of the efficiency of the process of gas exchange may be obtained by measuring the pulmonary transfer factor for carbon monoxide. This is assessed with an apparatus similar to that used for the helium-dilution technique for measuring lung volumes. Instead of using helium, which does not easily enter the blood, a known and very low concentration of carbon monoxide is used. The haemoglobin in the pulmonary capillaries very readily binds this gas. The patient inspires to TLC, holds the breath for 10 seconds, then expires fully. The difference between the inspired carbon monoxide concentration and the expired concentration is a measure of the efficiency of gas exchange and can be expressed per unit lung volume if TLC is simultaneously measured by the helium-dilution technique.

The bronchial provocation test, which is useful in the diagnosis of asthma, assesses airway hyperresponsiveness. It uses increasing concentrations of histamine or methacholine whilst measuring spirometry, FEV_1 and FVC and produces transient airflow limitation in susceptible patients.

Arterial blood sampling

In a sample of arterial blood, the partial pressures of oxygen (PaO_2) and of carbon dioxide ($PaCO_2$) and the pH can be measured. The arterial $PaCO_2$ will reflect the effective ventilation of alveoli that are adequately perfused with blood so that efficient gas exchange can take place. Provided the rate of carbon dioxide production by the body remains constant, the $PaCO_2$ will be directly related to the level of alveolar ventilation. The normal range is 4.7–6.0 kPa (36–45 mmHg). When alveolar ventilation is reduced, the $PaCO_2$ will rise. A number of different conditions may reduce alveolar ventilation. Alveolar ventilation rises and $PaCO_2$ may fall in response to metabolic acidosis, in very anxious individuals who hyperventilate and in many lung conditions that tend to reduce the oxygenation of the blood. The PaO_2 is normally in the range 11.3 to 14.0 kPa (80–100 mmHg). Any lung disease that interferes with gas exchange may reduce arterial PaO_2 (Box 12.18).

Box 12.18 Arterial blood gases

Type 1: Respiratory failure (on air)

- pH—7.43
- PCO_2—3.8
- PO_2—7.5
- HCO_3—22.0
- O_2 Sats—91%

Type 2: Decompensated respiratory failure (on air)

- pH—7.25
- PCO_2—9.3
- PO_2—7.5
- HCO_3—31.2
- O_2 Sats—92%

Type 2: Compensated respiratory failure (1 L O_2 and overnight bilevel positive airway pressure (BIPAP))

- pH—7.41
- PCO_2—6.3
- PO_2—8.3
- HCO_3—30.0
- O_2 Sats—94%

Exhaled nitric oxide

Nitric oxide (NO) is produced by the bronchial epithelium and its concentration is increased in asthma and other forms of airway inflammation; measurement can guide treatment in poorly controlled asthma.

Six-minute walking test

This validated test, which measures oxygen saturations, pulse rate and distance walked over a period of 6 minutes, is useful in measuring exercise tolerance in patients with chronic airway disease, heart failure and investigation of breathlessness.

Cardiopulmonary exercise testing (CPET)

A useful tool in the investigation of unexplained breathlessness, CPET provides an assessment of cardiopulmonary reserve. Together with cardiorespiratory and metabolic muscle function it measures oxygen consumption, CO_2 produced and ventilation whilst the patient uses a treadmill or a cycle ergometer.

Overnight oximetry and nocturnal polygraphy

Overnight oximetry measures the patient's O_2 saturations during sleep looking for evidence of

Box 12.19 Points to note when assessing the chest X-ray

- Name, hospital number and date of birth of patient and date (and time) of X-ray
- Position of the patient
- Outline of heart and mediastinum
- Position of the trachea
- Diaphragm
- Lung fields
- Bony skeleton

nocturnal hypoventilation and hypoxia. Nocturnal polygraphy is a multichannel sleep study that records pulse, O_2 saturations, nasal flow, body position and thoracic and abdominal wall movement and is used to investigate sleep disordered breathing.

Imaging the lung and chest

The chest X-ray

The chest X-ray is an important extension of the clinical examination (Box 12.19). This is particularly so in patients with respiratory symptoms. A normal X-ray taken some time before the development of symptoms should therefore not be accepted as a reason for not taking an up-to-date film. In many instances, it is of great value to have previous X-rays for comparison but, if these are lacking, then careful follow up with subsequent films may provide the necessary information.

The standard chest X-ray is a posteroanterior (PA) view taken with the film against the front of the patient's chest and the X-ray source 2 m behind the patient (see Fig. 12.5). The X-ray is examined systematically on a viewing box or computer screen, according to the following plan and referring to the thoracic anatomy described at the beginning of this chapter (see Figs 12.14–12.16 for more X-rays.)

The position of the patient

Is the patient straight or rotated? If straight, the inner ends of the clavicles will be equidistant from the midline of the vertebral body. This is important because any rotation usually tends to alter the appearance of the mediastinum and the hilar shadows.

The outline of the heart and the mediastinum

Is this normal in size, shape and position?

The position of the trachea

The position of the trachea is seen as a dark column representing the air within the trachea. Is the trachea centrally placed or deviated to either side? Is there any evidence of tracheal narrowing?

The diaphragm

Can the diaphragm be seen on each side? Is it normal in shape and position? Normally, the anterior end of the sixth or seventh rib crosses the mid-part of the diaphragm on each side, although the diaphragm on the right is usually a little higher than on the left. Are the cardiophrenic angles clearly seen?

The lung fields

For radiological purposes, the lung fields are divided into three zones:

1. The upper zone extends from the apex to a line drawn through the lower borders of the anterior ends of the second costal cartilages.
2. The mid-zone extends from this line to one drawn through the lower borders of the fourth costal cartilages.
3. The lower zone extends from this line to the bases of the lungs.

Each zone is systematically examined on both sides, and any area that appears abnormal is carefully compared with the corresponding area on the opposite side. The horizontal fissure, which separates the right upper and middle lobes, may sometimes be seen running horizontally in the third and fourth interspaces on the right side. Systematically compare not only the right and left lung fields, but also scan your eye up and down the chest film from top to bottom.

The bony skeleton

- Is the chest symmetrical?
- Is scoliosis present?
- Are the ribs unduly crowded or widely spaced in any area?
- Are cervical ribs present?
- Are any ribs eroded or absent?

As well as the standard PA view, lateral views are sometimes carried out to help localize any lesion that is seen. In examining a lateral view, as in Figure 12.9, follow this plan:

- Identify the sternum anteriorly and the vertebral bodies posteriorly. The cardiac shadow lies anteriorly and inferiorly.
- There should be a lucent (dark) area retrosternally that has approximately the same density as the area posterior to the heart and anterior to the vertebral bodies. Check for any difference between the two or for any discrete lesion in either area.
- Check for any collapsed vertebrae.The lowest vertebrae should appear darkest, becoming whiter as they progress superiorly. Interruption of this smooth gradation suggests an abnormality overlying the vertebral bodies involved.

The computed tomography (CT) scan

The routine chest X-ray consists of shadows from anatomical structures at all depths in the chest superimposed on one another. In computed tomography (CT) scanning, X-rays are passed through the body from many different positions around the body and capture on the other side, thereby recording the density of the tissues. The

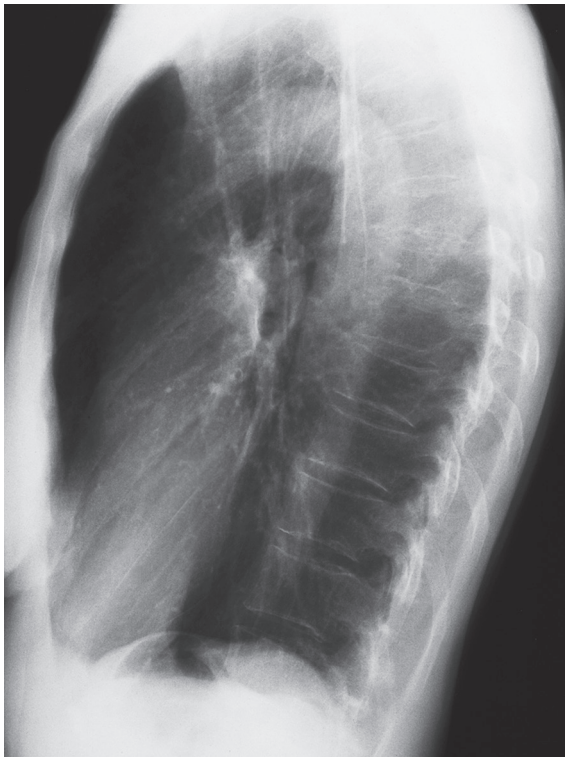


Figure 12.9 A lateral chest X-ray.

resulting information is processed by a computer to generate a series of cross-sectional images that display the various tissues in a slice dependent on their density. A thoracic CT scan thus is composed of a series of cross-sectional 'slices' through the thorax either at intervals or much more commonly as a contiguous set of images.

The CT scan plays a vital part in the imaging of thoracic pathology. Thin sections aid in the diagnoses of diffuse/interstitial lung disease and airways disease, such as bronchiectasis. The investigation, diagnosis and staging of lung cancer are heavily reliant on CT imaging. The addition of intravenous contrast that opacifies the pulmonary arteries allows detection of emboli as filling defects and has advantages over isotope lung scanning (see below) in diagnosing pulmonary embolism in patients with pre-existing lung disease.

Similarly, opacification of the aorta permits identification of vascular abnormalities, such as dissection, and cardiac gating enables the capture of slices in diastole only allowing accurate pictures of the coronary arteries to be obtained without the degrading influence of cardiac motion. In many cases this technique has replaced conventional coronary angiography. Using helical CT technology, scanners capture large volumes of data from which contiguous thin sections may be reconstructed, followed by post-processing techniques such as 3D reconstructions and the more clinically useful multi-planar reconstructions (Fig. 12.10).

Radioisotope imaging

For the lungs, the most widely used radioisotope technique is combined ventilation and perfusion scanning, used to aid the diagnosis of pulmonary embolism.

The perfusion scan is performed by injecting intravenously a small dose of macroaggregated human albumin particles labelled with technetium-99m (^{99m}Tc). A gamma-camera image is then built up of the radioactive particles impacted in the pulmonary vasculature; the distribution of perfusion in the lung can then be seen. The ventilation scan is obtained by inhalation of a radioactive gas, such as krypton-81m (^{81m}Kr), again using scanning to identify the distribution of the radioactivity.

Blood is usually diverted away from areas of the lung that are unventilated, so a matched defect on both the ventilation and perfusion scans usually indicates parenchymal lung disease. If areas of ventilated lung are not perfused (i.e. an unmatched defect), this is evidence in support of an embolism to the nonperfused area. Figure 12.11 shows a ventilation-perfusion isotope scan. The unmatched defects (areas ventilated by the inspired air but not perfused by blood) suggest a high probability of pulmonary embolism.

Magnetic resonance imaging

Magnetic resonance imaging (MRI) is useful in demonstrating mediastinal abnormalities and can help evaluate invasion of the mediastinum and chest wall by tumour. Apart from the fact that it does not use ionizing radiation, currently it has few other advantages over CT in imaging the thorax. MRI is particularly degraded by movement artefact in imaging the chest because of the relatively long data acquisition time and therefore is not used for assessing the lung parenchyma, but faster scanners are beginning to overcome this drawback.

Ultrasound

Ultrasound reveals much less detail than CT scanning but has the advantages that it does not involve radiation and, as it gives 'real-time' images, the operator can visualize what is happening as it happens. It is used for examining diaphragmatic movement and, when available, it is recommended that ward-based pleural procedures, such as chest drain insertion and pleural aspiration or biopsy, be undertaken under ultrasound guidance.

A paralysed hemidiaphragm usually results from damage to the phrenic nerve by a mediastinal tumour. If the patient is asked to make a sudden inspiratory effort (e.g. sniffing), the non-paralysed side of the diaphragm moves down, so intrathoracic pressure drops, and the paralysed side moves up.

Ultrasound is also valuable in distinguishing pleural thickening from pleural fluid. With real-time imaging, the latter can be seen to move with changes in posture. When such fluid is present, ultrasound

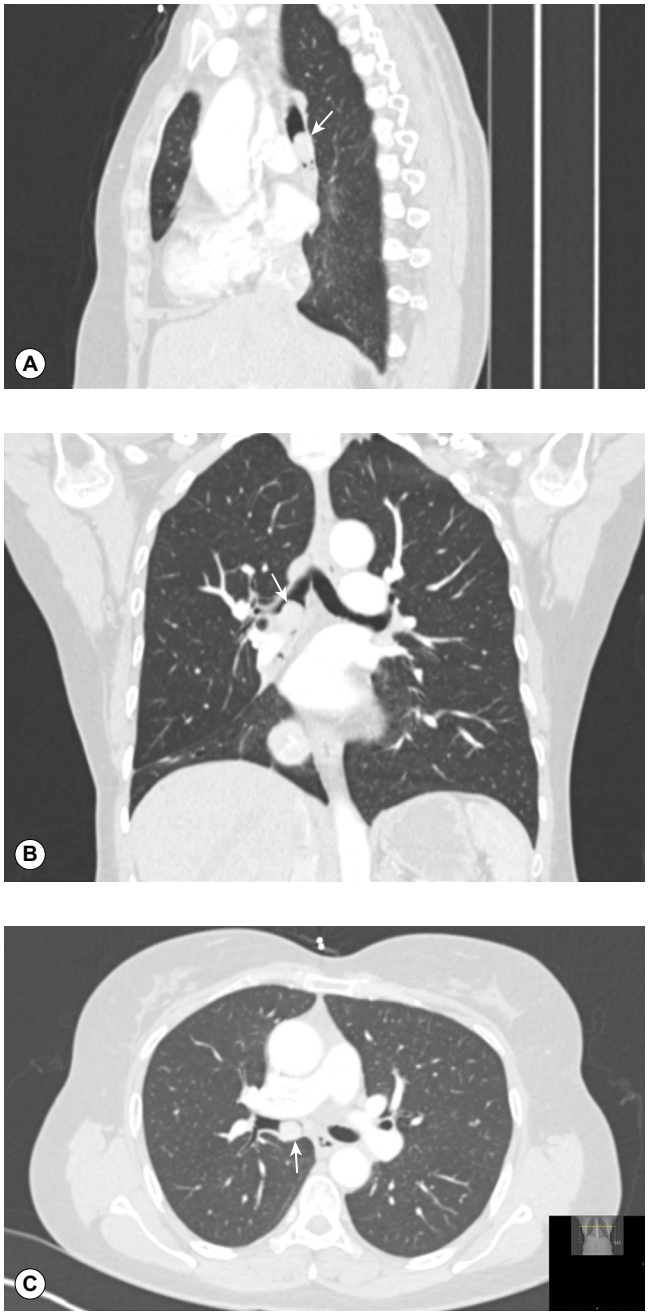


Figure 12.10 Axial images from a CT scan demonstrating an endobronchial lesion in the right main bronchus (**A**). The contiguous nature of the thin axial slices enables the display of the same structures in multiple planes. The coronal (**B**) and sagittal planes (**C**) have been created, allowing the surgeon to accurately place the endobronchial lesion anatomically to aid the planning of the operative resection (Courtesy of Dr Stephen Ellis.)

may be used to aid placement of a catheter to drain the collection and also to steer a draining catheter accurately into an intrapulmonary abscess.

Positron emission tomography (PET) scanning

In this technique, a radiolabelled 18-flurodeoxyglucose (FDG) molecule is administered, which is taken up by metabolically active tissues, such as cancers, showing as 'hot spots' on the image. It is useful in detecting regional and mediastinal lymphadenopathy and is widely used in the staging of lung cancers and to assess suitability for surgery in patients with lung cancer.

Flexible bronchoscopy and endobronchial ultrasound (EBUS)

Bronchoscopy is an essential tool in the investigation of many forms of respiratory disease. For discrete abnormalities, such as a mass seen on chest X-ray and suspected to be a lung cancer, bronchoscopy is usually indicated to investigate its nature. Under local anaesthesia, the flexible bronchoscope is passed through the nose, pharynx and larynx, down the trachea, and the bronchial tree is then inspected. Figure 12.12 shows a lung cancer seen down the bronchoscope. Flexible biopsy forceps, which are

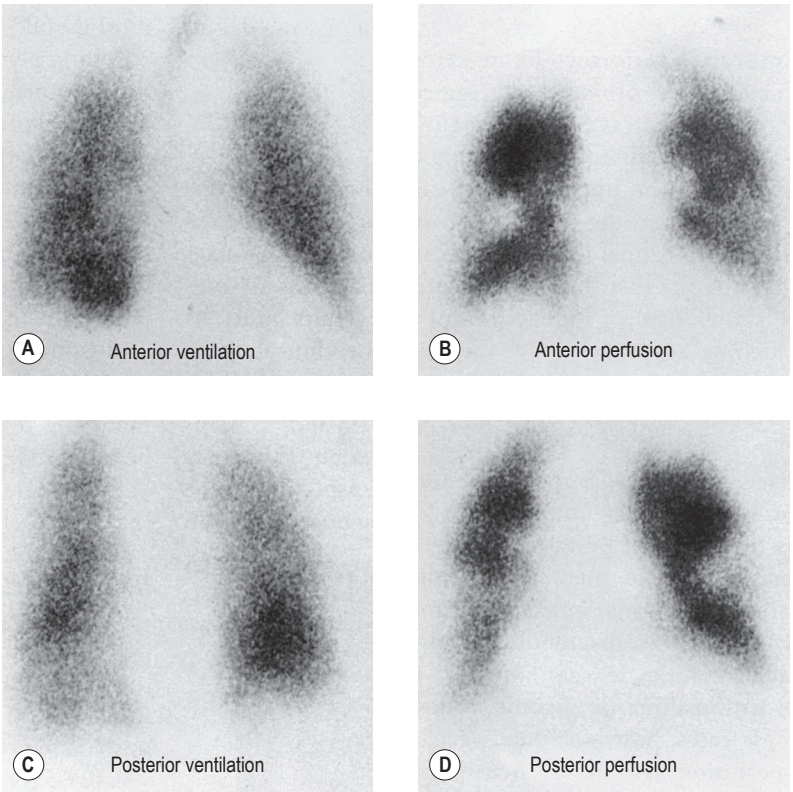


Figure 12.11 Ventilation/perfusion isotope scan of the lungs. Segmental and subsegmental loss of perfusion **(B and D)** can be seen with relatively normal ventilation **(A and C)**. The clear, punched-out areas in the perfusion **(B and D)** scans indicate areas of reduced isotope concentration during the perfusion scan. Thus, these are areas of reduced blood flow. The ventilation scans show normal aeration of the lungs as depicted by the isotope distribution in the pulmonary airways. These sequences of scans are suggestive of pulmonary embolism because they show impaired perfusion with normal ventilation.

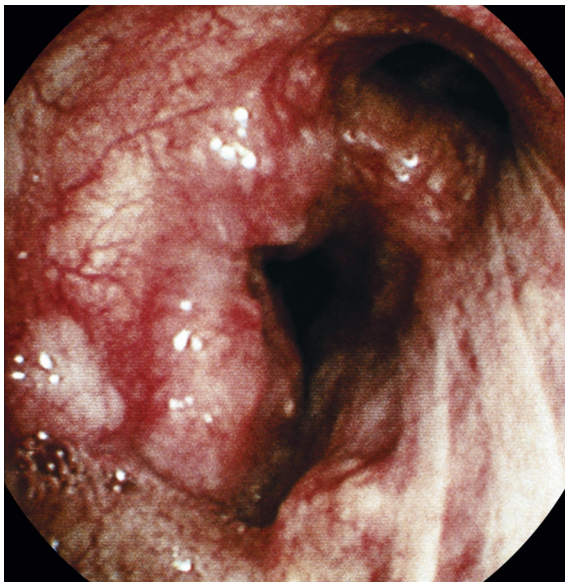


Figure 12.12 A lung cancer, seen down the bronchoscope.

passed down a channel inside the bronchoscope, are used to obtain tissue samples for histological examination. Similarly, aspirated bronchial secretions and brushings of any endobronchial abnormality can be sent to the laboratory for cytological examination.

At bronchoscopy, specimens are also taken for microbiological examination in order to determine the nature of any infecting organisms and should include samples for acid-fast bacilli (AFB). In diffuse interstitial lung disease, such as sarcoidosis or pulmonary fibrosis, the technique of transbronchial biopsy can be used to obtain small specimens of lung parenchyma for histological examination to help confirm the diagnosis.

Endobronchial ultrasound (EBUS) is gradually becoming more available. It involves a modified bronchoscope fitted with an ultrasound probe and a fine-gauge aspiration needle and is used to biopsy thoracic lymph nodes. The procedure is normally undertaken as a day case and under sedation. The scope is thicker than the average bronchoscope and is passed into the patient's airways via a plastic mouth guard rather than the nose. The ultrasound processor is able to image lymph nodes on the other side of the bronchial airways; the operator can then use the aspiration needle to puncture that bronchial wall and biopsy the lymph nodes. A similar procedure, endoscopic ultrasound (EUS), can be used via the oesophagus. Combining these two techniques allows all of the mediastinal lymph nodes to be biopsied. In the majority of cases, they have replaced mediastinoscopy as the biopsy technique of choice and are particularly useful in the diagnosis and staging of lung cancer, sarcoidosis and tuberculosis.

Navigational bronchoscopy is a developing specialist technique. It takes place with the anaesthetized patient lying on an electromagnetic table. This is linked to the patient's most up-to-date CT scan of the thorax via a computer using appropriate software to produce a 3D map of the thorax. It allows the operator, via a bronchoscope, to

guide a probe into a diseased area of lung, either for biopsy to make a diagnosis or to deliver treatment, such as microwave ablation directly to a tumour.

Pleural aspiration and biopsy

A pleural effusion (Fig. 12.15) can give rise to diagnostic problems and, sometimes, management problems when the amount of fluid causes respiratory embarrassment. When a pleural effusion is seen as a presenting feature in a middle-aged or older patient, the most likely cause is a malignancy. Less commonly, particularly in younger patients, it may be owing to tuberculosis. In either case, the diagnosis is best obtained by both aspiration of the fluid and pleural biopsy. Aspiration alone has a lower diagnostic yield.

After anaesthetizing the skin, subcutaneous tissues and pleura, pleural fluid may be aspirated by syringe and needle for microbiological and cytological examination. Large pleural effusions may need to be drained by an indwelling catheter, left in situ until the fluid has been fully removed. As noted above, ultrasound guidance can be helpful, particularly if the fluid is loculated in various pockets, and should be used whenever equipment and trained personnel are available.

Cytological examination of pleural fluid may demonstrate the presence of malignant cells. Many polymorphs may be seen if the effusion is secondary to an underlying pneumonic infection. With tuberculosis, the fluid usually contains many lymphocytes, although tubercle bacilli are rarely seen. Therefore, all pleural fluid samples should be cultured for possible tuberculosis, because this infection can coexist with other pathologies and it is so important not to miss it. In empyema, pus is present in the pleural cavity. It has a characteristic appearance and is full of white cells and organisms. An indwelling catheter should be left in situ to drain an empyema when possible.

The pleural fluid should also be examined for protein content. A transudate (resulting from cardiac or renal failure) can be distinguished from an exudate (from pleural inflammation or malignancy) by its lower protein content (<30 g/l). Light's criteria may also be applied (Box 12.20). When infection is suspected, pleural fluid pH should be measured in non-purulent effusions and, if not available, pleural fluid glucose should be assessed. A pH < 7.2 strongly suggests the need for pleural drainage (pleural fluid glucose <3.4 mmol/l). Pleural fluid lactate dehydrogenase (LDH) is also raised in the presence of infection.

It is recommended that pleural biopsies are undertaken under CT or ultrasound guidance, where available, and whilst Abram's pleural biopsy needles continue to be used in some centres, they are no longer the first choice of technique. Samples should be sent for histological examination, microscopy, culture and sensitivity (MC&S) and, whenever TB is a possibility, for AFB staining and TB culture.

Box 12.20 Light's criteria for diagnosing a pleural effusion

An effusion is exudative if it meets one of the following criteria:

- Pleural fluid protein/serum protein > 0.5
- Pleural fluid lactate dehydrogenase (LDH)/serum LDH ratio > 0.6
- Pleural fluid LDH > two-thirds the upper limit of normal serum LDH

Ridge thoracoscopy and video-assisted thoracoscopic surgery (VATS)

These techniques enable the pleural cavity to be examined directly and biopsies taken; VATS is now becoming the procedure of choice. The ridge method is normally performed under a general anaesthetic by a surgeon who uses direct vision down a rigid thoracoscope after the lung has been deflated. Increasingly, however, more minimally invasive procedures using flexible thoracoscopes attached to cameras are being used—VATS, and are not only able to biopsy the pleura but also biopsy the lung, mediastinal nodes and tumours, decortication of empysemas, lobectomy and pneumonectomy, pleurodesis and endoscopic stapled bullectomy (lung volume reduction surgery).

Lung biopsy

As noted above, the technique of transbronchial biopsy can be used to obtain samples of lung parenchyma, but often samples are too small for diagnosis. In this circumstance, biopsies of the lung taken at thoracoscopy may be of value. Occasionally, a formal open lung biopsy obtained at thoracotomy may be necessary.

When there is a discrete, localized lesion, it may be possible to obtain a biopsy percutaneously with the aid of CT scanning to direct the insertion of the biopsy needle (Fig. 12.13). All samples should be sent for histology, microbiology and TB culture.

Immunological tests

Asthma attacks may be caused by type I immediate hypersensitivity reactions on exposure to common environmental proteins known as allergens. In such individuals, an inherited tendency to produce exaggerated levels of immunoglobulin E (IgE) against these allergens is responsible. Part of the assessment of such allergic patients might include skin-prick tests (see Chapter 20). Alternatively, serum levels of specific (individual) IgEs against allergens may be measured by blood tests (formerly known as radioallergosorbent (RAST) tests) to demonstrate sensitization. The total IgE level is often raised in patients with asthma, rhinitis or eczema. Delayed (type IV, cell-mediated) hypersensitivity is shown by the Mantoux and Heaf skin tests, used to detect the presence of sensitivity to tuberculin protein.

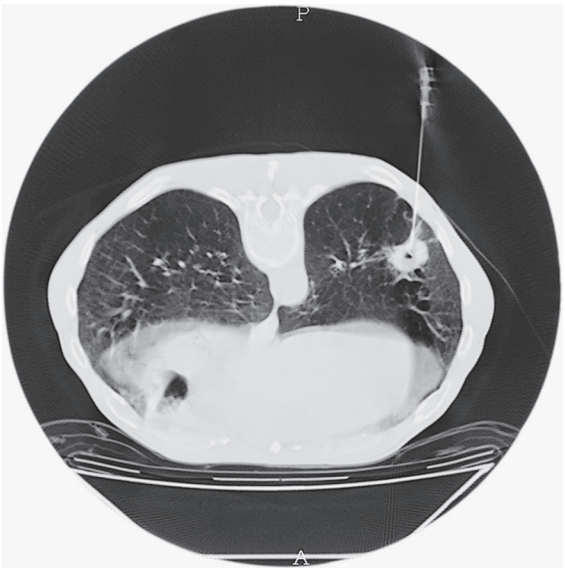


Figure 12.13 A CT-guided percutaneous biopsy in progress. The radiodense (*white*) structure penetrating the chest wall is the biopsy needle.

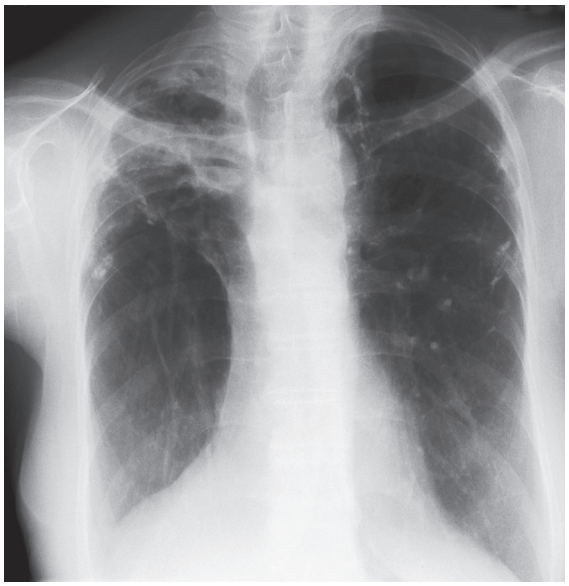


Figure 12.14 Chest X-ray showing right apical scarring and tracheal deviation (detectable clinically) from previous tuberculosis and hyperinflation of the lungs caused by chronic obstructive pulmonary disease in a 66-year-old long-term smoker with 5 years of increasing breathlessness.

Precipitating immunoglobulin G (IgG) antibodies in the circulating blood are present in patients with some fungal diseases, such as bronchopulmonary aspergillosis or aspergilloma. In patients suspected of having an allergic alveolitis, IgG antibodies may be demonstrated to the relevant antigens.

Tests for tuberculosis (TB)

Tuberculosis continues to be a worldwide problem, occurring most frequently as a pulmonary infection, but also commonly in the lymph nodes, as well as

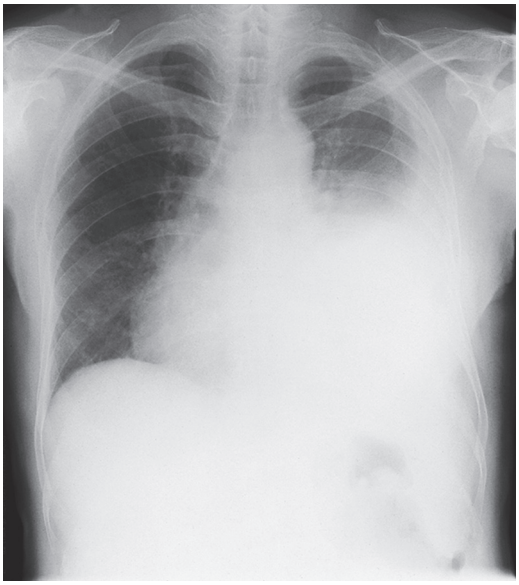


Figure 12.15 Chest X-ray showing a large left pleural effusion in a young man with a 4-month history of malaise, fever, night sweats and weight loss. The diagnosis of tuberculosis was confirmed on histology of a pleural biopsy and culture of the pleural fluid.

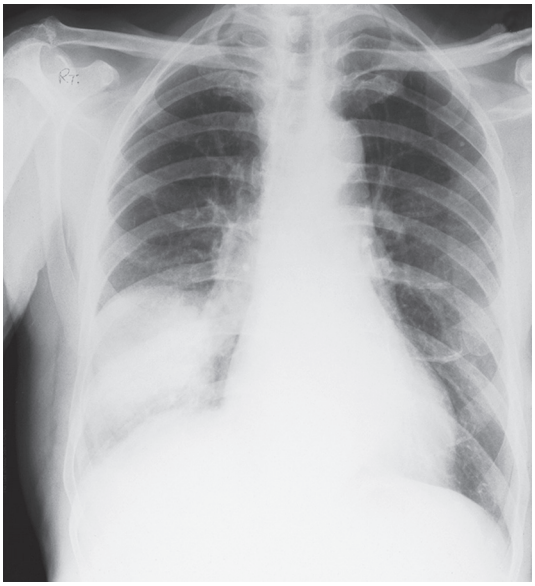


Figure 12.16 Chest X-ray showing a right basal pneumonia in a previously fit 40-year-old man with fever, breathlessness, central cyanosis and pleuritic pain. Chest signs included bronchial breathing and a pleural rub in the right lower zone. The cyanosis was caused by the shunting of deoxygenated blood through the consolidated lung, the increased respiratory rate leading to a low $PaCO_2$ because of increased clearance of carbon dioxide by the unaffected alveoli. *Streptococcus pneumoniae* was grown on blood cultures.

being able to affect any organ of the body. As outlined above, sending relevant samples for smear and culture is essential and often forgotten in hospitals where TB is less common. Sputum can easily be tested by light microscopy using a Zeihl-Neelsen or auramine stain to look for the AFB. Where available, culture

should be undertaken as drug monoresistance and multidrug-resistant TB (MDR TB) continue to be a major problem in the fight against the infection. Newer techniques help to diagnose active infection and drug resistance using molecular methods to detect *Mycobacterium tuberculosis* (MTB) complex DNA (e.g. the polymerase chain reaction (PCR)) assay Xpert® MTB/RIF (Cepheid, California, United States) and the line probe assay MTBDRplus® (Hain Lifescience, Nehren, Germany). More recently whole genome sequencing (WGS) is replacing routine culture and sensitivity testing to include different strains of mycobacterium and drug resistance.

Tests for latent TB and the Heaf and Mantoux skin tests are still widely used to look for evidence of previous TB exposure. In many centres, they are being superseded by blood tests that use the interferon gamma-releasing assay (IGRA), which measures interferon gamma released from T-cells activated by the presence of *Mycobacterium tuberculosis*. At the present time, the IGRA blood test does not differentiate between active and latent TB and should be used only to diagnosis latent disease. False-negative results can occur in disseminated and non-pulmonary active disease and can therefore be misleading when diagnosing active infection.